

Article

Impact of Board Committee Characteristics on Social Sustainability Reporting in Sub-Saharan Africa: The Moderating Role of Institutional Ownership

Marshall Wellington Blay ^{1,2}, Prosper Kweku Hoeyi ³ , Ebenezer Agyemang Badu ⁴ and Abdul Bashiru Jibril ^{5,*} 

¹ Faculty of Management Sciences, Department of Entrepreneurial Studies and Management, Durban University of Technology, Durban 4001, South Africa; marshall.blay@ttu.edu.gh

² Department of Accounting & Finance, Takoradi Technical University, Takoradi P.O. Box 256, Ghana

³ Department of Business Support Studies, Central University of Technology, Bloemfontein 9301, South Africa; phoeyi@cut.ac.za

⁴ Department of Agricultural Economics and Extension Education, Akenten Appiah-Menka University of Skills Training & Entrepreneurial Development, Asante-Mampong P.O. Box 40, Ghana; ebagyemang@aamusted.edu.gh

⁵ School of Management and Economics, University of Kurdistan Hewlêr, 30 Metre Avenue, Erbil 44001, Iraq

* Correspondence: mallambash13@gmail.com

Abstract: The corporate strategic planning of businesses in sub-Saharan Africa (SSA) largely focuses on immediate financial performance with minimal credence to social sustainability. Thus, studies on the linkage between corporate governance (CG) and sustainability reporting have focused on developed economies. This study therefore investigated the role of institutional ownership in the impact of board committee characteristics on social sustainability reporting. This study involved strongly balanced panel data with 1969 observations of 275 publicly listed non-financial firms in SSA within the timeframe of 2012 to 2021. Data were analyzed using STATA 14.1. The hypotheses were tested using the two-step system of the generalized method of moment (GMM) using the Arellano–Bond dynamic panel data estimation method. The rate of social sustainability reporting was 39.4%. Relatively, Mauritian and South African firms had the most effective board committee characteristics and higher levels of social sustainability reporting. Although institutional ownership had no significant effect on social sustainability reporting, it moderated the effect of sustainability committee independence and sustainability committee gender diversity on social sustainability reporting. This paper presents a new perspective on the corporate governance and social sustainability literature by examining the effect of institutional ownership on board committee characteristics and social sustainability reporting in SSA. In terms of policy implication, there is the need for mandatory regulatory and legal CG framework that is regularly updated at national and regional levels in SSA to motivate listed firms to establish sustainability committees with efficient characteristics to promote social sustainability reporting.

Keywords: corporate governance; board committee; social sustainability reporting; sub-Saharan Africa (SSA)



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1. Introduction

In many parts of the world, there is an increasing demand for sustainability reporting among stakeholders due to climate change, changing consumer preferences, and environmental accidents (Dienes and Velte 2016). There is maximum growth in issues of sustainability reporting in the US, Europe, and Asia (Matta 2017). However, in the developing world, especially in sub-Saharan Africa, participation in sustainability reporting is a recent phenomenon (Addjin-Tettey et al. 2021). In every industry, sustainability has emerged as more and more crucial for businesses. Sixty-two percent of business leaders believe that a sustainability strategy is essential for competitiveness today, and another

22% believe that it will be in the years to come (Haanaes 2016). In a survey conducted by McKinsey in February 2010, although companies actively managing sustainability are reaping the benefit of superior shared value, most companies fail to manage sustainability actively (Krechovská and Procházková 2014). In a nutshell, sustainability is a business strategy that considers an organization's operations in the ecological, social, and economic spheres in order to create long-term value. The foundation of sustainability is the idea that creating these kinds of tactics encourages the continued existence of businesses. Although sustainability reporting is persistently becoming prevalent in the developed world, its practice and embracement are still extremely low in developing countries, especially among businesses in sub-Saharan Africa (Marquis and Qian 2014; Wachira and Mathuva 2022). The corporate strategic planning of businesses in SSA largely focuses on immediate financial performance with minimal credence to sustainability.

There are few studies on corporate governance and sustainability reporting in Sub-Saharan Africa (SSA), largely because of the low level of sustainability reporting among enterprises in the region as a result of the underdeveloped capital market. As a result, the majority of research on corporate governance and sustainability focuses mostly on companies in Western, industrialized countries like the USA, Canada, the UK, and Australia (Tkachenko et al. 2020; Dienes and Velte 2016). It is not unexpected that these studies concentrate on industrialized countries, since an increasing number of governments in Europe and North America require sustainability reporting, particularly for certain kinds or sizes of firms. In Asia, developed or recently developed nations like China (Marquis and Qian 2014), Japan (Fukukawa and Moon 2004), and Pakistan (Sharif and Rashid 2014) have been the focus of the majority of sustainable development reporting research studies. Also, notwithstanding the increasing importance of sustainability in every aspect of human life and all sectors of the global economy, studies in sub-Sahara Africa on corporate governance have largely emphasized its relationship with firm performance with limited emphasis on sustainability (Tilt et al. 2021). During their examination of sub-Saharan Africa's company sustainability reporting situation, Tilt et al. (2021) reported a struggling state.

Although the continent is not sequestered from the global environmental challenges of climate change, there are still ineffective national and regional level policies to confront both social and environmental issues (Tilt et al. 2021). Evidence on the level of sustainability reporting in SSA has also been very scarce to inform policy effective building and implementation due to the difficulty in obtaining firm-level information. Thus, studies on sustainability and corporate social responsibility reporting are confronted with enormous difficulty due to the poor political and socioeconomic conditions of the region (Kühn et al. 2018). Thus, the extent to which the existing social and environmental regulatory frameworks like SDGs, Africa's Development Agenda -Agenda 2063, the GRI, and the International Integrated Reporting Council (IIRC) have contributed to addressing the environmental and social challenges of the region is still not clear. More so, there is limited sustainability reporting regulatory framework in SSA (Bartels et al. 2016). Although there are thirteen SSA countries that are signatories to the Sustainable Stock Exchange (SSE) initiative (Tilt et al. 2021), mandatory reporting is a requirement for listing in only four countries: Zimbabwe, Nigeria, Namibia, and South Africa (Tilt et al. 2021).

As the leading country in SSA with effective and regularly updated regulatory framework, South Africa has primarily been the focal point of studies on sustainability reporting in SSA (Ahmed Haji and Anifowose 2016). Evidence on the state of sustainability reporting among many SSA countries is therefore unavailable. Nonetheless, the available evidence indicates that CSR in the region is largely community-based, and primarily designed to promote community development and assist in poverty alleviation to strengthen the community-level relationship with companies (Hamidu et al. 2016). In order to strengthen the community-level relationship, and enhance stakeholder engagement to strengthen legitimacy, listed businesses in the sub-region are more likely to report information on corporate social responsibility than environmental information (Johnson-Rokosu and Olanrewaju 2016; Lauwo et al. 2016). Nonetheless, the general level of investment of companies and

information disclosure on CSR is still limited and underdeveloped based on report from KPMG 2017 (cf. Baldarelli et al. 2017), due to significant reporting barriers, such as institutional, economic, technical, and cultural challenges. Many countries in the sub-region lack robust regulations and guidelines for sustainability reporting. This results in inconsistent and non-comprehensive reporting practices.

Institutional characteristics and ownership structure are also recognized as potential barriers to effective CG practices and sustainability reporting in SSA. With higher institutional concentration limiting control to few large investors, a small number of large investors dominating corporate decisions in Africa often prioritize short-term financial gains over long-term sustainability goals and robust governance practices, which consequently limit sustainability reporting. This therefore indicates that the level of institutional ownership concentration of publicly listed businesses in SSA is important for development of efficient national- and regional-level sustainability reporting regulatory and legal frameworks. However, there is also the argument that institutional ownership has been vital in business practice, regulations, higher interest in sustainability reporting, and research since the 2008–2009 financial crisis (Faller and zu Knyphausen-Aufseß 2018). With the perceived greater experience and resources of institutional block holdings, their level of influence on corporate strategies is relatively greater than the other forms of block holdings (Klettner 2021). Since the institutions are more cognizant of the public interest, they monitor the investment corporations' boards of directors and put pressure on leadership to step up corporate sustainability initiatives (Basse Mama and Mandaroux 2022). According to Kordsachia et al. (2021), the majority of sustainable investors who have signed the United Nations Principles for Responsible Investment (PRI) or comparable voluntary networks are institutional investors. Thus, in comparison to other kinds of equity ownership, organizational block holdings typically require more corporate sustainability information and efficient leadership tools, and they are likely to put pressure on management to improve their sustainability performance.

However, apart from the restricted focus of earlier research on the mitigating function of ownership arrangements, limited emphasis is particularly accorded to the role of institutional ownership concentration as boundary condition in the relationship between board committee characteristics and sustainability disclosure, particularly in sub-Saharan Africa. Thus, the specific objectives of this paper are to examine level of social sustainability reporting among listed non-financial firms in SSA, determine how the board committee characteristics of the firms influence their level of social sustainability reporting, and further determine whether the level of institutional ownership concentration of the firms serve as a boundary condition in the relationship between the board committee characteristics and the level of social sustainability reporting. This research is important because it clarifies the necessity of sustainable information disclosure on the ownership and corporate governance structures in SSA.

Sustainability Regulatory Framework in Sub-Saharan Africa

The sustainability reporting regulatory framework in sub-Saharan Africa is gradually developing as countries in the region recognize the importance of transparency and accountability in environmental, social, and governance (ESG) issues. While the framework is not as advanced or widespread as in some other regions, several countries and regional initiatives are making significant strides. At the global level, the region is also guided by the Global Reporting Initiative (GRI) and Integrated Reporting (IR). Many companies in sub-Saharan Africa are adopting the GRI standards for sustainability reporting. These standards provide a comprehensive framework for reporting on a wide range of sustainability issues. Some organizations are also adopting integrated reporting frameworks, which combine financial and non-financial data to provide a holistic view of an organization's performance.

At the country-specific level, some countries like South Africa, Mauritius, Ghana, Kenya, and Nigeria have developed a sustainability framework to guide sustainability reporting. As a leader in sustainability reporting in sub-Saharan Africa, South Africa

requires listed companies to produce integrated reports. The Johannesburg Stock Exchange (JSE) mandates that companies adhere to the King IV Report on Corporate Governance, which includes guidelines on sustainability reporting. The Nigerian Stock Exchange (NSE) has introduced sustainability disclosure guidelines for listed companies, encouraging them to report on ESG factors. The recently published Nigeria Code of Corporate Governance 2018 (NCCG Code) was published by the FRCN on 15 January 2019 and seeks to promote public awareness of essential corporate values and ethical practices by recommending practices and principles that affected companies are to adhere to (Fagbemi et al. 2022). The Nigerian Code of Corporate Governance 2018 (the NCCG Code or the Code), which cuts across all sectors of the Nigerian economy, was issued by the Financial Reporting Council of Nigeria (FRCN) to replace all existing sectoral codes (Nwosu et al. 2021). These guidelines are currently voluntary but aim to promote best practices in sustainability reporting. In Kenya, the Nairobi Securities Exchange (NSE) has launched sustainability reporting guidelines for listed companies, promoting transparency in ESG issues. These guidelines are aligned with international standards like the GRI and the UN Sustainable Development Goals (SDGs). In Ghana, the Companies Act 2019 is the principal law that governs the operations of companies and sets out the fundamental corporate governance requirements for companies. It is also guided by the Section 209 of the Securities Industry Act 2016 (Act 929); this Code was issued by the Securities and Exchange Commission on the 8th of October 2020.

Besides country-specific regulatory frameworks, there are also regional initiatives that aim to promote sustainability disclosure among listed public companies in Africa. The African Securities Exchanges Association (ASEA) is working to harmonize sustainability reporting standards across African stock exchanges. This initiative aims to create a more consistent and transparent reporting environment in the region. Also, the African Integrated Reporting Council (AIRC) promotes integrated reporting across the continent, encouraging companies to provide comprehensive reports that cover financial and non-financial performance.

With the exception of South Africa, the regulatory framework of the sub-Saharan African region is largely voluntary and disrupted by several challenges. South Africa is a notable leader, with mandatory integrated reporting for listed companies, while other countries are developing voluntary guidelines and promoting best practices. Many of the countries in the region lack the regulatory frameworks and institutional capacity to enforce mandatory sustainability reporting. Strengthening these institutions is crucial for effective implementation and monitoring.

2. Theoretical Underpinning and Hypothesis Development

The agency theory and the legitimacy theory are two of the main theories that guide this research. The agency hypothesis, which Jensen and Meckling (1976) established, proposed that the conflicting interests among a company's directors, shareholders, and key debt financing sources form the basis of a business's governance. According to agency theory, companies represent their shareholders' interests. In other words, when shareholders invest in corporate ownership, they are entrusting the administration of the corporation's directors and officers with their resources (Kumar et al. 2022). In larger companies, the interests of officers and stockholders in the short and long terms can diverge dramatically. The short-term need for profits and the information asymmetry between officers, directors, and shareholders are the main causes of this. Officers and directors may become disengaged from shareholder interests, and it is believed that this divide in interests affects their decisions and actions. Thus, it is believed that the agency conflict between management and shareholders results from the decentralization of ownership as well as authority in the open financial system. According to the agency theory, corporate governance can lessen the principal–agent dilemma by establishing effective board committees with efficient and effective qualities. This could then result in a higher standard of sustainability reporting. Based on the agency theory, effective corporate governance mechanisms, such as

well-structured board committees, are crucial to align the interests of management with those of shareholders. Nonetheless, institutional ownership can play a moderating role in this relationship through enhanced monitoring mechanisms, demand for accountability, and mediating conflicts of interest between managers and shareholders, advocating for practices that align with long-term shareholder value, which often includes robust social sustainability initiatives. However, if institutional investors prioritize short-term financial returns, this can weaken the positive impact of board committees on sustainability reporting, as the focus shifts away from long-term social sustainability goals. This therefore suggests that the level of institutional ownership concentration could serve as a boundary condition in the relationship between the board characteristics of publicly listed firms and the level of social sustainability reporting in SSA.

Additionally, as the legitimacy theory emphasizes, corporate social responsibility carried out at the community level can help establish legitimacy, which in turn helps improve sustainability reporting. Within the subject of sustainability reporting research, legitimacy theory is one of the most well-established theoretical lenses of study (Montecchia et al. 2016; Gavana et al. 2016). In the literature on social and environmental accounting, it has been the most often used theoretical framework for analyzing the disclosure of social and environmental information (Crane and Glozer 2016). Although there is considerable diversity in the theoretical viewpoints, experts generally agree that the primary incentive for firms to reveal social information is the pursuit of legitimacy (Montecchia et al. 2016). Suchman (1995) provided the most often cited and likely most common definition of legitimacy, which is defined as a generalized perception or presumptions that an entity's actions are desirable, proper, or appropriate within a socially constructed set of norms, values, beliefs, and definitions. The idea that a company should act in a way that society considers socially acceptable and that it is required to express such behavior is one way that the legitimacy theory lens of analysis explains corporate social responsibility (CSR) and the reporting of it.

With the legitimacy theory suggesting that companies seek to operate within the bounds of societal norms and values to maintain legitimacy and ensure their survival, social sustainability reporting is clearly an essential strategic tool for publicly listed companies in SSA to demonstrate their legitimacy by showing their commitment to social and environmental responsibilities. Particularly, institutional investors, especially those with a strong commitment to ESG criteria, can pressure companies to enhance their social sustainability reporting. This pressure reinforces the board committees' efforts to adopt practices that are viewed favorably by society in order to safeguard their reputation and the reputation of their portfolio companies. Institutional investors, especially those operating globally, are often attuned to international standards and societal expectations regarding corporate responsibility. Their involvement can lead to the adoption of best practices in social sustainability reporting, as board committees align their policies to meet these expectations. Also, institutional ownership can influence the composition and independence of board committees by advocating for the inclusion of independent directors with expertise in sustainability, which could eventually enhance the effectiveness of these committees in overseeing social sustainability initiatives. Institutional investors can also advocate for the training and development of board committee members in social sustainability issues, which would enhance the quality of sustainability reporting as committees become more knowledgeable and effective. In line with the legitimacy theory, it can also be inferred that institutional ownership could potentially serve as a boundary condition in the relationship between board committee characteristics and social sustainability reporting in SSA. Generally, the extent of the influence of institutional ownership structure depends on the investors' commitment to long-term sustainability goals versus short-term financial returns.

2.1. Hypotheses Development

2.1.1. Board Committee Characteristics and Sustainability Reporting

According to Boone et al. (2007), board committees are now a more official and regulated part of the board of directors in the corporate sector. In order to ensure that

management decisions align with shareholder interests, the board of directors is expected to carry out oversight and monitoring responsibilities (Alhossini et al. 2021). Certain responsibilities, like risk management, audits, and compensation, may call for specialized knowledge. Boards began forming committees to assess key aspects of the businesses they run in this environment. Following the passage of the Sarbanes–Oxley Act (SOX) in 2002, the major stock exchanges, the NASDAQ and NYSE, required that companies establish compensation, audit, nomination, governance, as well as compensation boards in addition to social, investment, executive officer, and financial committees (Bansal and Singh 2022). Nonetheless, it has been noted that established board committee qualities are important for the disclosure of sustainability (Subramaniam et al. 2017). Board committees should have no more than four members, as this is considered the optimal number for effective operation (Agyemang et al. 2020). Additionally, it is advised that the audit committee have more than 50% of independent directors in order to strengthen the committee’s independence (Saeed et al. 2022). A number of studies in the body of research on corporate governance have found a strong correlation between sustainability and the characteristics of the board committee (Ame et al. 2017). The board committees under discussion in this study’s context were the audit and CSR/ESG committees. There has been discussion and conjecture on the attributes of these committees, including their size, independence, meetings, and gender diversity.

2.1.2. Board Committee Size and Sustainability Reporting

In general, a board committee of four members is considered sufficient for good sustainability reporting performance (Anyigbah et al. 2023). The existing literature acknowledges that the size of board committees influences the efficacy of the committees’ functions in terms of sustainability reporting (Okere et al. 2021). Agyemang et al. (2020) found that the size of a company’s board committees had a beneficial impact on social sustainability reporting. Rabi (2021) and Kumari et al. (2022) found that board committee size had a beneficial influence on sustainability disclosure. A larger board committee size is often interpreted as implying higher duty allocation, variety, and a lower level of burden, all of which improve stakeholder representation (Jizi et al. 2013). Thus, the larger the size of board committees, the higher the level of social sustainability reporting (Kumari et al. 2022). As a result, a larger board committee size is more representative of stakeholders (Jizi et al. 2016) and more aware and effective in meeting social duties (Kumari et al. 2022). As a result, higher board committee sizes are likely to be more closely related to social sustainability reporting. Based on this context, the following is hypothesized:

H1. *Board committee size is positively and significantly associated with SSR.*

2.1.3. Board Committee Independence and Sustainability Reporting

It is generally known that having a larger proportion of independent directors effectively improves sustainability reporting by requiring a higher degree of monitoring management (Liao et al. 2015). The amount of pressure on management to disclose sustainability rises exponentially when there is a greater percentage of independent directors on board committees (Shamil et al. 2014). It is common knowledge that independent directors are knowledgeable individuals who can oversee management, monitor it, and offer sage counsel and recommendations about social transparency (Khairiddine et al. 2020). According to Almaqtari et al. (2022), the body of existing literature indicates that board committee independence has a beneficial impact on sustainability reporting. A substantial favorable association between board independence and social reporting was also reported by Aliyu (2019). Greater board independence helps a company become more socially sustainable, and social performance and board independence have a strong and favorable relationship (Ortiz-de-Mandojana et al. 2016). Additionally, a number of studies have shown a positive and significant correlation between the performance of social and corporate sustainability and the proportion of independent directors on board committees (Kumari et al. 2022).

Independent directors are more likely than other members of the board committees to enforce a higher level of sustainability reporting (Ammer et al. 2020). Accordingly, social sustainability reporting is more likely to be reported when board committees have higher representation from independent board directors. This study therefore hypothesizes the following:

H2. *Board committee independence is positively and significantly associated with SSR.*

2.1.4. Board Meetings and Sustainability Reporting

Effectiveness is indicated by a board committee meeting four times a year on average (Perego and Kolk 2012). Advocates of the agency theory viewpoint contend that increased board meeting frequency is linked to improved oversight and could, thus, have a favorable impact on the companies' disclosure of sustainability (Shamil et al. 2014). A key metric for assessing the level of executive oversight and board activity is the frequency of meetings). Regular board meetings are seen to improve the board's engagement in company operations and motivate managers to take into account the interests of all parties involved, not just shareholders). There is a claim that regular board meetings enhance supervision duties, which could have an effect on the caliber of corporate reporting (Karamanou and Vafeas 2005). According to Jizi et al. (2013), there is general agreement that holding regular board meetings improves coordination, communication, and agency expenses. Notwithstanding the reported importance of board committee meetings to the general performances of organizations, there is scant empirical literature on the linkage between board committee meetings and sustainability reporting, especially in sub-Saharan Africa. However, limited studies have largely reported a positive effect of board committee meetings on social sustainability reporting (Bansal and Singh 2022). In the study of 92 Indian software companies from 2011 to 2018, Bansal and Singh (2022) reported that board meetings increase the level of social sustainability reporting. Based on the reviewed literature, this study hypothesizes the following:

H3. *Board committee meetings positively and significantly affect SSR.*

2.1.5. Gender Diversity and Sustainability Reporting

Gender diversity on boards is a notion that has garnered attention from academics and corporations alike, as it is believed to improve the effectiveness of boards in good governance. Additionally, research examining the correlation between gender diversity and reporting on sustainability revealed a favorable link (Katmon et al. 2019). According to this research, gender diversity has a major impact on sustainability disclosures, demonstrating the value of female directors to companies (Tilt et al. 2021). Because women's perspectives differ from men's, gender diversity therefore promotes balanced decision-making (Bakar et al. 2019). Moreover, women are known to support rational decisions that could enhance the sustainability strategies and, as a result, the sustainability reporting of the organizations (Al-Shaer and Zaman 2016; Bakar et al. 2019). According to Al-Shaer and Zaman (2016), female directors exhibit better sensitivity to sustainability concerns, generosity towards problems in the community, and stakeholder orientation, as they make greater consideration for the environment. According to research by Al-Shaer and Zaman (2016), gender diversity on boards improves social sustainability reporting in the United Kingdom. According to earlier research, there is a strong positive correlation between board gender diversity and social sustainability reporting (Harjoto et al. 2015; Ibrahim and Hanefah 2016). This study makes the following assumptions based on the body of existing literature demonstrating the relevance of female directors from the standpoint of legitimacy theory:

H4. *Board gender diversity in committees positively and significantly affects SSR.*

2.1.6. Institutional Ownership as a Moderator

The percentage of a company's accessible shares held by endowments, mutual or pension funds, insurers, investment firms, private foundations, or other sizable organizations that look after other people's money is known as institutional ownership (Ismail et al. 2020). Numerous scholars have examined institutional investors' function as watchdogs over corporations. The reason for this is the high expense of monitoring; only substantial shareholders, such institutional investors, can profit sufficiently to warrant the need for monitoring (Grossman and Hart 1980). Compared to board members, who might have little to no capital invested in the company, significant shareholders might have greater motivation to keep an eye on managers. In addition, big institutional investors possess the chance, means, and capacity to oversee, control, and sway managers (Shleifer and Vishny 1986). According to Smith (1996), Del Guercio and Hawkins (1999), and other researchers, managers may become more focused on the company's accomplishments as a result of institutional investors' monitoring of the company, rather than on opportunistic or self-interested behavior. An institutional owner may be able to keep a close eye on management and push them to reveal additional information, including social information, according to the agency theory (Ntim et al. 2013). The more influential qualities of an institutional owner have an impact on the board's decision-making from a social standpoint, as any opposition to such a perspective may destroy opportunities for firms to invest and increase operating expenses, as demonstrated by the BP oil spill in the Gulf of Mexico in 2010 and the Exxon oil spill in 1989 (De Villiers et al. 2011). According to Faller and zu Knyphausen-Aufseß (2018), institutional investors have the ability to select directors who possess experience and a strong resource base. These directors will be more receptive to the organization's strategic choices about its social policies and strategies. The conversations thus imply that the degree of institutional ownership in a company determines the extent to which board committee attributes impact the quality of social sustainability reporting (ESR). As a result, this research suggests the following:

H5. *IO moderates the link between board committee characteristics and SSR.*

3. Materials and Methods

3.1. Dataset and Source

The study population comprises all non-financial firms listed on the stock exchanges of sub-Saharan African countries as of 31 December 2021. The inherent nature of the data gives rise to a panel data framework, characterized by both time series and cross-sectional dimensions. Out of the total 29 stock exchanges in Africa, 15 are located in sub-Saharan Africa. Notwithstanding these many stock exchanges in sub-Saharan Africa, a limited number are matured and part of the African Securities Exchanges Association (ASEA), including the Nairobi Securities Exchange (Kenya), Stock Exchange of Mauritius (Mauritius), Nigerian Exchange Group (Nigeria), Johannesburg Stock Exchange Limited (South Africa), Zimbabwe Stock Exchange (Zimbabwe), Ghana Stock Exchange (Ghana), Botswana Stock Exchange (Botswana), and Malawi Stock Exchange. However, not all these stock exchanges had data for the study period of 2012 to 2021.

The final dataset used in this study comprised manually extracted listed companies from Nigeria, Ghana, Kenya, South Africa, Zimbabwe, and Mauritius as of 31 December 2021. These six countries are an adequate representation of the eight countries with matured stock exchanges. The selection of the six countries and the years selected were driven by the availability of the requisite data. There were three critical conditions that the firms met before they were included in the sample: First, the firm should have issued a report and have all the required data from 2012–2021. Second, the firm should have been listed on the stock exchange of the selected countries. Third, the issued report should be written in the English language. Firms that did not meet the above criteria were excluded from the sample. Finally, 197 firms from the six selected countries were included in this study.

3.2. Model Specification

Multiple periods across a multiplicity of firms provided characterize this paper. Thus, this paper relies on paneled data for the analysis. This form of data permits the modeling of variations in the behavior of different firms over time. The general regression model defining the relationship between the independent variable, the assumed moderator, control variables (firm-specific characteristics), and the dependent is shown in Equation (1).

$$Y_{it} = \beta_0 + \sum_{n=1}^k \beta X_{it} + \sum_{n=1}^k \beta v_{it} + \sum_{n=1}^k \beta \psi_{it} + \mu_{it} \quad (1)$$

where

Y_{it}	= Social sustainability reporting of the i th firm at time period t ;
β_0	= Intercept;
X_{it}	= Firm-specific characteristics of i th firm at time period t ;
v_{it}	= Board committee characteristics of i th firm at time period t ;
Ψ_{it}	= Institutional ownership of i th firm at time period t ;
B	= Coefficient of the independent variables;
μ_{it}	= Error or the disturbance term;
$n = 1, \dots, k$	= From the first variable to the k th variable;
$i = 1, 2, 3, \dots, N$	= Firm index or the cross-sectional dimension;
$t = 1, 2, 3, \dots, N$	= Times series dimension.

This study employed the generalized moment method (GMM) estimation technique. This method is designed for data with a ‘small’ T , and large N panels (Phillips 2019), a condition met by the data of this study as there are few periods ($T = 10$) and many individuals ($N = 667$ listed non-financial firms). Due to their correlation with potential current and historical errors, the independent variables are not strictly exogenous. Resolving the fixed individual effect problems in the existing data also requires addressing the large number of individual-specific factors. In panel data, there are additional issues with autocorrelation and heteroscedasticity inside individuals but not between them, hence the need to employ a more robust method like GMM estimation. The GMM is built on two sets of equations. These sets of equations are Equation (2) (original) and Equation (3) (transformed). The GMM system uses both the first-difference transformed equation and the level equation.

$$SSR_{it} = \delta SSR_{it-1} + \beta_1 MBV_{it} + \beta_2 DTA_{it} + \beta_3 AUDSize_{it} + \beta_4 AUCIndep_{it} + \beta_5 AUDMeet_{it} + \beta_6 Csrsize_{it} + \beta_7 Scind_{it} + \beta_8 Csr bmt_{it} + \beta_9 Scgd_{it} + \beta_{10} INST_{it} + \sigma_i + \mu_{it} \quad (2)$$

The deduced original equation (2), or level, is assumed to be a random walk model with a persistent dependent variable. Thus, Equation (2) is expressed in level form with first differences (FDs) as instruments. In this equation, the introduced lag dependent (SSR_{it-1}) is assumed to correlate with the fixed effect (σ_i), or the unobserved specific individual characteristics and the error term (μ_{it}). Individual-specific patterns of heteroskedasticity and serial correlation may be present in the idiosyncratic disturbances (those that are not related to the fixed effects) (Roodman 2009). The problem of correlation between the fixed effect and the lag dependent is resolved through the first-difference GMM. However, considering the evaluation of the moderation concept, Equation (3) was developed in line with the first differentiation.

$$\begin{aligned} SSR_{it} - SSR_{it-1} = & SSR_{it-1} - SSR_{it-2} + \beta_1 MBV_{it} - MBV_{it-1} + \beta_2 DTA_{it} - DTA_{it-1} \\ & + \beta_3 AUDSize_{it} - AUDSize_{it-1} + \beta_4 AUCIndep_{it} - AUCIndep_{it-1} + \\ & \beta_5 AUDMeet_{it} - AUDMeet_{it-1} + \beta_6 Csrsize_{it} - Csrsize_{it-1} + \beta_7 Scind_{it} - Scind_{it-1} \\ & + \beta_8 Csr bmt_{it} - Csr bmt_{it-1} + \beta_9 Scgd_{it} - Scgd_{it-1} + \beta_{10} INST_{it} - INST_{it-1} + \sigma_i - \sigma_{i-1} + Interactions + \mu_{it} - \mu_{it-1} \end{aligned} \quad (3)$$

The first differencing is achieved by transforming Equation (2), the original. Consequently, levels are used as instruments in the FD form of Equation (3). Hence, the system GMM uses more instruments than the FD GMM. The differencing eliminates the fixed

effect (σ_i), as this component does not vary over time. In estimating, unlike in the FD GMM, the differential and level equations are both used by the GMM system. Heteroscedasticity and serial autocorrelation are present when the Windmeijer standard error option is used (Windmeijer 2005). Also, the standard system GMM estimator employs both the levels and differenced data. Instead, if and just if the instrument's requirement in the theorem is met, system GMM estimates can use the levels data and forward orthogonal deviations (Phillips 2019). Notwithstanding the elimination of the fixed effect, due to the lag in the dependent variable, there is still a chance that it will be endogenous. (SSR_{it-1}) in Equation (3) in the term $SSR_{it-1} = SSR_{it-1} - SSR_{it-2}$ could correlate with the μ_{it} in the term $\Delta\mu_{it} = \mu_{it} - \mu_{it-1}$. The predefined variables in Equation (3), which are not necessarily exogenous, have the potential to become endogenous due to their potential relationship with μ_{it-1} . Hence, unlike the mean deviation transformation, larger lags of the regressors stay orthogonal to the error and available as instruments (Arellano and Bover 1995).

It subtracts the means of all upcoming accessible observations of a variable, as an alternative to subtracting the previous observation from the concurrent observations. Because it is computable for all observations, with the exception of the last for each individual, regardless of the number of gaps, this second transformation minimizes data loss. Lagged observations are confirmed as instruments but do not enter the equation in forward orthogonal transformation. A number of GMM tests in dynamic data models are carried out to guarantee effective and reliable estimators of the system GMM. Among these tests are the Arellano–Bond test of serial correlation; the Sargan/Hansen test of over-identification restrictions; and the differences in the Sargan/Hansen test of exogeneity (Roodman 2009). The first serial correlation tests the appropriateness of the data for the dynamic model, whereas the second serial correlation tests the goodness of the lag dependent as an instrument. A test for the validity of the instruments is the Sargan/Hansen test of over-identification constraints. Checking for exogenous subsets of instruments in the level's equations is another function of the Sargan/Hansen test of exogeneity. These steps are necessary to justify adopting the system GMM and the 2SLS estimation methods.

In terms of the measurement of the variables, the dependent variable, sustainability reporting, was measured using the number of indicators reported by each company in the social indicator category according to GRI guidelines. With regard to the independent variables, audit committee size was measured as the total directors and non-directors in the audit committee, audit committee independence was measured as the number of non-directors and non-executive directors in the audit committee divided by audit committee members size (%), audit committee meeting is the number of meetings held by the audit committee of board of companies per annum, sustainability committee size was measured as the total directors and non-directors in the sustainability committee, sustainability committee independence was measured as the number of non-directors and non-executive directors in the risk committee divided by sustainability committee members size (%), sustainability committee meetings was measured as number of meetings held by the sustainability committee members in a year, and sustainability committee female diversity was measured as the number of female risk committee members divided by sustainability committee members size (%).

4. Results

4.1. Board Committee Characteristics

Table 1 shows that the average size of the audit committees (Acsiz) of the listed SSA businesses was 4.6 members. Listed businesses in Nigeria have the highest audit committee size of 5.5 members whereas Ghanaian listed businesses have the lowest audit committee size of 3.7 members. The only country with listed firms that failed to adhere to the recommended audit committee size of 4 members was Nigeria. With a 95.1 percent average, the listed businesses in Mauritius have the highest audit committee independence (Acind), and Nigerian listed businesses with 52.2 percent have the lowest audit committee independence. The audit committees were characteristically independent, as about 81

percent of the members were independent directors. The average annual number of meetings of the audit committees of the listed SSA businesses was four times. Listed businesses in Mauritius have the highest audit committee meetings (Acmt) of 4.8 times annually relative to the lowest of 3.5 times in Ghana.

Table 1. Board committee characteristics of sub-Saharan Africa.

Country	Audsize	Aucindep (%)	Audmeet	Csrsz	Scind (%)	Scgd (%)	Csrbmt
Ghana	3.7 (1.1)	90.7 (14.4)	3.5 (.7)	0	0	0	0
Kenya	4.3 (1.3)	95.6 (12.4)	4.4 (1.8)	0.0 (0.5)	0.6 (7.3)	0.3 (2.9)	0.0 (0.4)
Mauritius	3.8 (1.2)	95.1 (14.4)	4.8 (1.4)	0.1 (4.0)	0.5 (4.0)	0.0	0.0 (0.2)
Nigeria	5.5 (1.0)	52.2 (19.2)	3.9 (0.9)	0.0	0.0	0.0	0.0
South Africa	4.4 (1.7)	90.6 (19.0)	3.9 (1.5)	3.5(2.1)	46.5 (34.6)	18.8 (22.3)	2.0 (1.5)
Zimbabwe	3.9 (1.2)	88.8 (17.9)	3.6 (1.2)	0.3 (1.1)	1.8 (9.1)	5.8 (22.9)	0.1 (0.5)
Total	4. 6 (1.5)	81.2 (24.9)	4.0 (1.3)	1.6 (2.2)	20.8 (32.6)	8.9 (18.8)	0.9 (1.3)

Variables: audit committees (Acsz), audit committee independence (Acind), audit committee meetings (Acmt), CSR/ESG sustainability board committee (Csrsz), sustainability committee (Scind), sustainability committee gender diversity (Scgd), and CSR/ESG sustainability board meeting (Csrbmt). Source: Extracted from [Blay \(2024\)](#).

With listed businesses in Ghana and Nigeria missing social sustainability committees, not much can be said about the female presentation. However, in the case of listed businesses in South Africa, Zimbabwe, and Kenya, the female representation on the sustainability committees of listed businesses in these countries were 18.8%, 5.8%, and 0.3%, respectively. In Mauritius, although the publicly listed companies have established sustainability committees, there are no female representations on the committees. The higher representation of two females in ten with sustainability committee membership among listed South African businesses could be attributed to the mandatory requirements of the King IV Report on Corporate Governance.

Table 1 demonstrates that social sustainability and corporate social responsibility governance committees are absent from listed companies in Ghana and Nigeria. In South Africa, there is a greater prevalence of listed companies with social sustainability and CSR committees. This finding is not surprising since both Nigeria and Ghana do not have a well-structured legal and regulatory framework to guide and enforce corporate governance practices, contrary to the situation in South Africa, which is guided by the King IV Report on Corporate Governance.

4.2. Social Sustainability Reporting of Sub-Saharan African Businesses

Table 2 shows that the percentage of all social measurement items like community, health and safety, donation and gift, data protection and privacy, human rights, customer and complaints, educational sponsorship, public health sponsorship, and others disclosed or reported by sub-Saharan African businesses listed on the stock market is about 39 percent. The listed businesses that reported the highest proportion of their social disclosure measured items were Mauritius and South Africa. With 51.3 percent and 49 percent of social sustainability disclosure in Mauritius and South Africa, respectively, these two countries have the highest level of social sustainability disclosure practices relative to Ghana, Kenya, and Nigeria. The high level of social sustainability reporting in these countries could be attributed to the mandatory measures instituted in these countries. For instance, the higher sustainability disclosure of South African listed firms could be attributed to the instituted mandatory reporting measure in the form of the King Reports on Corporate Governance and the B-BEE legislation in the country ([Wachira and Berndt 2019](#)). Although all the sub-Saharan African countries considered in this study have some form of regulation that both explicitly and implicitly encourage the issuance of sustainability disclosures, there are variations in the application of the regulations ([Wachira and Mathuva 2022](#)). There is evidence of the high level of mandatory sustainability reporting demands in South Africa and Mauritius relative to the other countries ([Wachira and Berndt 2017](#)).

Table 2. Social sustainability reporting.

Country	Mean (%)	Std. Dev.
Ghana	21.9	26.2
Kenya	26.6	18.9
Mauritius	51.3	17.5
Nigeria	30.9	15.8
South Africa	49.0	23.5
Zimbabwe	30.2	25.5
Total	39.4	23.6

Note: Results are mean/averages plus or minus standard deviations. Source: Extracted from [Blay \(2024\)](#).

4.3. Correlational Analysis

The results of the intervariable correlation are presented in Table 3. Table 3 shows that both the market-to-book value (mbv) ($r = 0.114$, $p < 0.05$) and debt-to-asset ratio (bta) ($r = 0.112$, $p < 0.05$) positively and significantly correlated with the social sustainability disclosure of the listed non-financial firms. The audit committee characteristics of the listed firms, such as audit committee size ($r = 0.113$, $p < 0.05$), audit committee independence ($r = 0.193$, $p < 0.05$), and audit committee meeting ($r = 0.176$, $p < 0.05$), were positive and significant predictors of the social sustainability disclosure of the listed firms. The characteristics of the sustainability committees of the boards of the listed non-financial firms in SSA, such as CSR committee size ($r = 0.417$, $p < 0.05$), sustainability committee independence ($r = 0.388$, $p < 0.05$), sustainability committee female gender diversity ($r = 0.360$, $p < 0.05$), and sustainability committee meetings ($r = 0.395$, $p < 0.05$) significantly and positively correlated with the social sustainability disclosure of the firms. The institutional ownership of the listed firms also positively and significantly correlated with the social sustainability reporting of the firms.

Table 3. Intervariable correlation.

Variables	1	2	3	4	5	6	7	8	9	10	11
1 Mbv	1.000										
2 Dta	0.034	1.000									
3 audsize	−0.004	0.146 *	1.000								
4 audindep	0.069 *	−0.173 *	−0.637 *	1.000							
5 audmeet	−0.125 *	0.057 *	0.130 *	0.048 *	1.000						
6 csrsize	0.127 *	−0.110 *	−0.105 *	0.287 *	0.008	1.000					
7 Scind	0.111 *	−0.086 *	−0.121 *	0.284 *	0.008	0.495 *	1.000				
8 Scgd	0.091 *	−0.043 *	−0.081 *	0.239 *	0.033	0.336 *	0.757 *	1.000			
9 csrbmt	0.130 *	−0.079 *	−0.102 *	0.269 *	0.054 *	0.499 *	0.370 *	0.424 *	1.000		
10 blkinsown	0.014	0.004	0.025	−0.020	−0.009	0.042 *	0.024	0.041 *	0.031	1.000	
11 sdisclos	0.114 *	0.112 *	0.113 *	0.193 *	0.176 *	0.417 *	0.388 *	0.360 *	0.395 *	−0.001	1.000

* signifies statistical significance at 0.05.

4.4. Descriptive Statistics

The descriptive statistics of the variables utilized in this paper are presented across all countries in Table 4. The table therefore provides information on the characteristics of the audit committee and the sustainability committee. The committee characteristics emphasized size or committee membership of audit committee (audsize) or sustainability committee (csrsize), independence or the percentage representation of non-executive members on audit committees (Aucindep) or sustainability committees (Scind), the total number of meetings organized annually by audit committees (audmeet) or sustainability committees (csrbmt), and sustainability committee female gender diversity (scgd). Further descriptive statistics are also provided for institutional ownership concentration (Blkinsown), social sustainability reporting or disclosure (Sdisclos), market-to-book value (mbv), and debt-to-asset ratio (dta).

Table 4. Descriptive Statistics Across Countries.

Country		Mbv (%)	Dta (%)	Audsize	Aucindep (%)	Audmeet	Csrsz	Scind (%)	Scgd (%)	Csrbmt	blkins-n (%)	Sdisclos (%)
Ghana	Min	−526.6	4.9	2.0	33.3	2.0	0.0	0.0	0.0	0.0	0.0	0.0
	Max	818.5	143.3	7.0	100.0	5.0	0.0	0.0	0.0	0.0	95.0	100.0
	Mean	7.0	71.6	3.7	90.7	3.5	0.0	0.0	0.0	0.0	46.5	21.9
	SD	83.8	29.7	1.0	14.5	0.7	0.0	0.0	0.0	0.0	23.7	26.2
	Range	1345.1	138.4	5.0	66.7	3.0	0.0	0.0	0.0	0.0	95.0	100.0
Kenya	Min	−20.8	4.8	2.0	25.0	1.0	0.0	0.0	0.0	0.0	0.0	0.0
	Max	33.4	180.7	11.0	100.0	15.0	6.0	83.3	33.3	4.0	98.0	75.0
	Mean	2.1	50.8	4.3	95.6	4.4	0.0	0.6	0.3	0.0	44.9	26.6
	SD	4.7	25.6	1.3	12.4	1.8	0.5	7.3	2.9	0.4	22.9	18.9
	Range	54.2	175.9	9.0	75.0	14.0	6.0	83.3	33.3	4.0	98.0	75.0
Mauritius	Min	0.0	9.8	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
	Max	171.9	102.4	8.0	100.0	9.0	3.0	33.3	0.0	2.0	95.0	92.9
	Mean	4.0	47.2	3.8	95.1	4.8	0.1	0.5	0.0	0.0	47.2	51.3
	SD	19.5	18.1	1.2	14.4	1.4	0.4	4.0	0.0	0.2	25.3	17.5
	Range	171.9	92.5	8.0	100.0	9.0	3.0	33.3	0.0	2.0	95.0	92.9
Nigeria	Min	−1176.2	3.6	2.0	0.0	1.0	0.0	0.0	0.0	0.0	0.0	0.0
	Max	103.9	395.5	9.0	120.0	10.0	0.0	0.0	0.0	0.0	98.0	100.0
	Mean	1.3	68.5	5.5	52.2	3.9	0.0	0.0	0.0	0.0	51.1	30.9
	SD	42.8	42.8	1.0	19.2	0.9	0.0	0.0	0.0	0.0	26.5	15.8
	Range	1280.1	391.9	7.0	120.0	9.0	0.0	0.0	0.0	0.0	98.0	100.0
South Africa	Min	−7.9	−46.8	2.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
	Max	816.5	15,648.2	13.0	100.0	12.0	11.0	114.3	100.0	9.0	444.0	100.0
	Mean	6.3	84.1	4.4	90.6	3.9	3.5	46.5	18.8	2.0	50.4	49.0
	SD	26.8	661.5	1.7	19.0	1.5	2.1	34.6	22.3	1.5	26.5	23.5
	Range	824.4	15,695.0	11.0	100.0	12.0	11.0	114.3	100.0	9.0	444.0	100.0
Zimbabwe	Min	−1201.7	2.3	2.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
	Max	371.6	206.7	9.0	100.0	12.0	6.0	66.7	100.0	3.0	95.0	92.9
	Mean	0.1	54.7	3.9	88.8	3.6	0.3	1.8	5.8	0.1	45.4	30.2
	SD	98.9	30.2	1.2	17.9	1.2	1.1	9.1	22.9	0.5	26.7	25.5
	Range	1573.3	204.4	7.0	100.0	12.0	6.0	66.7	100.0	3.0	95.0	92.9
Total	Min	−1201.7	−46.8	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
	Max	818.5	15,648.2	13.0	120.0	15.0	11.0	114.3	100.0	9.0	444.0	100.0
	Mean	3.9	71.1	4.6	81.2	4.0	1.6	20.8	8.9	0.9	49.2	39.4
	SD	46.4	439.3	1.5	25.0	1.4	2.3	32.6	18.9	1.4	26.1	23.6
	Range	2020.2	15,695.0	13.0	120.0	15.0	11.0	114.3	100.0	9.0	444.0	100.0

4.5. Board Committee Characteristics and Social Sustainability Reporting

The GMM model was utilized to test the developed hypotheses related to board committee characteristics, institutional ownership, and social sustainability disclosure of sub-Saharan African businesses listed on the stock market. The considered dependent variable was social sustainability disclosure. The independent variable and the moderating variable were board committee characteristics and institutional ownership, respectively. To test the developed hypotheses, the two-step system of the generalized method of moment (GMM) using `xtabond2` in STATA 14.1. Strongly balanced panel data with 1969 observations of 275 groups within the timeframe of 2012 to 2021 were modeled using the Arellano–Bond dynamic panel data estimation method. In the estimated model, the utilized instrumental variables included market-to-book value (Mbv) and debt-to-asset (DTA) ratio. The GMM type or conditions of estimation were no-levels, no-diffsargan, robust, two-step, and small. The results of the Arellano–Bond dynamic panel data estimation two-step system GMM are presented in Table 5. The hierarchical regression modeling method involving three models was utilized in testing the moderation concept.

Model 1 in Table 5 demonstrates that the audit committee size (AUDSize) is positively and significantly correlated with the degree of social sustainability disclosure ($\beta = 1.078$, $p < 0.05$) in the absence of ownership considerations for the listed enterprises. Therefore, there is a correlation between the extent of sustainability disclosure or reporting and the membership size of audit committees of publicly traded companies in sub-Saharan Africa.

Also, without the consideration of ownership of the listed businesses, audit committee independence (AUCIndep) is positively and significantly associated with the level of social sustainability disclosure ($\beta = 0.067, p < 0.05$). Thus, a higher degree of independence of the audit committees of publicly listed businesses in sub-Saharan Africa is associated with increasing level of sustainability disclosure or reporting. With no particular consideration to ownership of the listed businesses, the amount of social sustainability disclosure is positively and strongly correlated with the number of audit committee meetings (AUDMeet) held each year ($\beta = 0.944, p < 0.01$). Thus, an increasing number of meetings held by the audit committees of publicly listed businesses in sub-Saharan Africa is associated with an increasing level of sustainability disclosure or reporting.

Table 5. Dynamic panel data estimation, two-step difference GMM.

Social	Model 1	Model 2	Model 3
Controls			
Social (L1.)	0.436 (0.043) ***	0.428 (0.044) ***	0.416 (0.043) ***
Mbv	0.002 (0.005)	0.002 (0.005)	0.001 (0.005)
Dta	0.010 (0.023)	0.005 (0.023)	0.007 (0.022)
Independent			
AUDSize	1.078 (0.416) **	1.207 (0.437) ***	2.492 (1.23) **
AUCIndep	0.067 (0.029) **	0.078 (0.031) **	0.097 (0.075)
AUDMeet	0.944 (0.285) ***	1.169 (0.417) ***	1.615 (1.51)
Csrsz	0.354 (0.981)	0.406 (0.970)	2.347 (1.19) **
Scind	0.009 (0.035)	0.009 (0.036)	−0.182 (0.082) **
Scgd	0.007 (0.054)	0.008 (0.053)	0.333 (0.146) **
Csrbmt	−0.246 (0.834)	−0.201 (0.825)	0.657 (1.58)
Moderator			
Blkinsown		0.001 (0.037)	0.303 (0.242)
Interactions			
AUDsize × INST			−0.027 (0.024)
AUCIndep × INST			−0.001 (0.002)
AUDMeet × INST			−0.007 (0.029)
csrsz × INST			−0.041 (0.026)
scind × INST			0.004 (0.002) **
scgd × INST			−0.006 (0.003) **
csrbmt × INST			−0.023 (0.038)
Number of obs	1957	1934	1934
Number of instruments	370	370	370
F-statistics	13.66	10.56	9.45
Prob > F	0.000	0.000	0.000
AR(2) in first difference	$z = 2.86, p = 0.004$	$z = 2.89, p = 0.004$	$z = 2.82, p = 0.005$
Sargan test	$\chi^2 = 408, p = 0.041$	$\chi^2 = 413, p = 0.024$	$\chi^2 = 389, p = 0.086$

Note: Standard errors in the parentheses; *** and ** denote significance at 1%(0.001) and 5%(0.05), respectively. Control variables: Market-to-book value (Mbv) and debt-to-asset ratio (Dta); independent variables: audit committees (Acsz), audit committee independence (Acind), audit committee meetings (Acmt), CSR/ESG sustainability board committee (Csrsz), sustainability committee (Scind), sustainability committee gender diversity (Scgd), CSR/ESG sustainability board meeting (Csrbmt); moderating variables: block institutional ownership (Blkinsown); dependent variable: social sustainability disclosure (Sodisclos). Source: Extracted from [Blay \(2024\)](#).

Model 2 of Table 5 further reveals that the audit committee features of publicly traded institutional block holding companies are positively and significantly related to the amount of social sustainability disclosure. Model 2 indicates a positive and substantial correlation ($\beta = 1.207, p < 0.01$) between institutional block holding firms' audit committee size (AUDSize) and their level of social sustainability disclosure. The degree of social sustainability disclosure is positively and strongly correlated with the audit committee independence (AUCIndep) of institutional block holding companies ($\beta = 0.078, p < 0.05$). Additionally, there is a positive and substantial correlation ($\beta = 1.167, p < 0.01$) between the total number of audit committee meetings (AUDMeet) held by institutional block holding corporations and the level of social sustainability disclosure. Model 2 makes clear that companies classified as institutional block holders have a comparatively larger marginal influence on sustainability reporting or disclosure due to the features of the audit committees of publicly traded companies.

Additionally, Model 3 of Table 5 demonstrates that the audit committee size (AUD Size) is positively and significantly correlated with the degree of social sustainability disclosure in the presence of the moderator and interactions ($\beta = 2.492, p < 0.05$). Additionally, Model 3 demonstrates a positive and substantial correlation between the size of the CSR/ESG sustainability board committee (Csrsize) and the degree of social sustainability disclosure ($\beta = 2.347, p < 0.05$). Model 3 further demonstrates a positive and substantial correlation ($\beta = 0.333, p < 0.05$) between the level of social sustainability disclosure and the gender diversity of the sustainability committee (Scgd). Model 3 reveals that the amount of social sustainability disclosure is significantly and adversely correlated with the independence of the sustainability committee (Scind) in the presence of the moderator and interactions ($\beta = -0.182, p < 0.05$). Model 3 shows that institutional block holdings significantly and positively moderate the effect of sustainability committee independence (Scind) on the level of social sustainability disclosure ($\beta = 0.004, p < 0.05$). Also, institutional block holdings significantly and negatively moderate the effect of sustainability committee gender diversity (Scgd) on the level of social sustainability disclosure ($\beta = -0.006, p < 0.05$).

4.6. Robustness Check

Four distinct models are provided in Table 6 for both ordinary least square (OLS) and fixed-effect estimation methods for the purpose of checking the robustness of the base system GMM model. The econometric lapses of these two methods for estimating dynamic panel models is evident in the unstable nature of some of the coefficients of both the OLS and the fixed-effect methods. More so, there was evidence of heteroscedasticity and autocorrelation issues regarding the estimated OLS models, indicating the inadequacies of the usage of this estimation method in analyzing dynamic panel models. Also, these estimation methods failed to take into consideration the lagged dependent variable. Comparatively, the base system GMM model addressed these lapses in estimation methods and provided more stable and robust coefficients to accurately estimate the effect of board committee characteristics on the social sustainability reporting practices of listed non-financial firms in sub-Saharan Africa.

Table 6. OLS and fixed-effect estimation methods as robustness check.

Social	Fixed-Effect			OLS		
	Model 1	Model 2	Model 3	Model 1	Model 2	Model 3
Controls	2.031 (0.288) ***	2.665 (0.333) ***	3.066 (0.307) ***	3.445 (2.854)	11.159 (3.106) ***	
Mbv	−0.005 (0.006)	−0.005 (0.006)	−0.005 (0.006)	0.014 (0.011)	0.007 (0.010)	0.006 (0.010)
Dta	−0.001 (0.001)	−0.007 (0.013)	−0.010 (0.013)	−0.002 (0.001)	−0.088 (0.015) ***	−0.085 (0.015) ***
Independent						
AUDSize	0.137 (0.225)	0.202 (0.229)	0.275 (0.234)	1.808 (0.326) ***	1.745 (0.327) ***	1.937 (0.339) ***
AUCIndep	0.049 (0.016) ***	0.058 (0.016) ***	0.064 (0.016) ***	0.143 (0.028) ***	0.118 (0.021) ***	0.122 (0.022) ***
AUDMeet	0.272 (0.241)	0.211 (0.249)	0.156 (0.254)	2.588 (0.312) ***	2.504 (0.315) ***	2.413 (0.320) ***
Csrsize	1.161 (0.273) ***	1.132 (0.277) ***	0.052 (0.506)	2.134 (0.332) ***	2.196 (0.332) ***	2.762 (0.721) ***
Scind	−0.034 (0.017) **	−0.038 (0.018) **	0.026 (0.033)	0.072 (0.022) ***	0.043 (0.022) **	−0.050 (0.050)
Scgd	0.055 (0.023) **	0.055 (0.023) **	0.079 (0.053)	0.064 (0.028) **	0.077 (0.028) ***	0.195 (0.074) ***
Csrbmt	0.296 (0.385)	0.338 (0.385)	0.302 (0.388)	0.632 (0.529)	1.114 (0.528) **	1.266 (0.535) **
Moderator						
Blkinsown		−0.023 (0.012) *	−0.022 (0.017)		−0.015 (0.015)	−0.096 (0.056) *
Interactions						
AUDsize × INST			−0.099 (0.212)			0.001 (0.006)
AUCIndep × INST			0.007 (0.016)			0.001 (0.001)
AUDMeet × INST			−0.199 (0.205)			0.009 (0.005) *
csrsize × INST			0.021 (0.009) ***			−0.010 (0.012)
scind × INST			−0.001 (0.001) ***			0.002 (0.001) *
scgd × INST			−0.001 (0.001)			−0.002 (0.001)
csrbmt × INST						10.471 (3.235) ***
R2 within	0.0211	0.0243	0.0293			
R2 between	0.2273	0.2324	0.2153			
R2 overall	0.1702	0.1780	0.1678	0.2114	0.2286	0.2337

Note: * is the significant level of at $p < 0.05$, ** $p \leq 0.01$ and *** $p \leq 0.001$.

5. Discussion and Implications

Although there is increasing concern and stakeholder interest in sustainability reporting, the level of sustainability reporting as envisaged in the extant literature is still limited. Thus, the evidence from this study shows that the social sustainability reporting level of 39.4 percent in sub-Saharan African businesses is still low since it is far below the 53.5 percent in developed countries (Bhatia and Makkar 2020). In developed countries, sustainability reporting is mandatory and required by regulations and the legislature (Wachira and Mathuva 2022). South Africa and Mauritius seemingly stand above the rest of the sub-Saharan African countries in terms of social sustainability reporting due to the mandatory requirement, and the existence of strong regulations in these countries like in developed countries (Wachira and Mathuva 2022). According to Molo (2014), listed South African corporations are required by the Johannesburg Stock Exchange (JSE) to generate an integrated report that combines environmental, social, and economic disclosures into a single, all-inclusive document. However, the practice of sustainability reporting is optional and not standardized or controlled at the national level in other nations, such as Ghana, Nigeria, Zimbabwe, and Kenya (Wachira and Berndt 2019), and as a result, it is relatively low. Aside from the fact that sustainability reporting is optional in most of sub-Saharan Africa, many publicly traded companies in the region have inadequately characterized board committees because many of them do not even have sustainability committees at all.

The characterized audit and sustainability committee sizes of the publicly listed businesses were positively and significantly associated with social sustainability reporting. These findings suggest that the effective and efficient size of both audit and sustainability committees of publicly listed businesses is associated with a higher level of social sustainability reporting. These results corroborate the hypothesized (H1) favorable and noteworthy impact of board committee size on social sustainability reporting in sub-Saharan Africa. This result confirms previous research showing that board committee size affects how well committees do their duties in terms of sustainability reporting (Okere et al. 2021). The size of the board committees of companies was reported in the study of Agyemang et al. (2020) to positively influence social sustainability reporting. The research conducted by Rabi (2021) and Kumari et al. (2022) underscored the beneficial impact of board committee size on sustainability disclosure. The main explanation is that a larger board committee size results in a higher degree of duty allocation and variety, and a lower level of burden, all of which improve stakeholder participation (Jizi et al. 2013). Consequently, a larger board committee size is more effective and mindful of its social duties (Kumari et al. 2022). It is also more representative of stakeholders (Jizi et al. 2016). Social sustainability reporting was favorably and strongly correlated with the gender diversity of the sustainability committees of the listed companies. This implies that increasing gender diversity in corporate social sustainability committees of the listed businesses is associated with an increasing level of social sustainability reporting. This finding supports the hypothesis (H4) that board committees' gender diversity positively and significantly affects social sustainability reporting.

Numerous studies in the body of existing literature corroborate this conclusion, commenting similarly on the favorable and noteworthy impacts of gender diversity on board committees on social sustainability reporting (Katmon et al. 2019). According to Tilt et al. (2021), these studies demonstrate the importance of female directors in the corporate world. Because women's perspectives differ from men's, gender diversity therefore promotes balanced decision-making (Bakar et al. 2019). Furthermore, women have a reputation for endorsing sensible choices that might improve the companies' sustainability strategies and, consequently, sustainability reporting (Al-Shaer and Zaman 2016; Bakar et al. 2019). However, social sustainability reporting was adversely and significantly correlated with the corporate social responsibility committee's independence. This suggests that a higher level of independence in a corporate social sustainability committee is associated with a decreasing level of social sustainability reporting. This finding therefore contradicts the hypothesis (H2) that board committee independence is positively and significantly associated with social sustainability reporting. This result also runs counter to the body

of research that shows board committee independence has a favorable and substantial impact on social sustainability reporting (Almaqtari et al. 2022). According to this research, corporate social sustainability committees' strategic and business decisions are not always impacted by the participation of independent directors.

The discussions suggest that some major characteristics of the board committees of institutional block holding firms listed in sub-Saharan Africa significantly matter to the degree of social sustainability reporting. Although the characteristics of firms identified as institutional block holdings do not significantly matter, the institutional block holding characteristics of the firms moderated the significant effect of corporate social sustainability committee independence and gender diversity on social sustainability reporting in sub-Saharan Africa. This suggests that in larger institutional investment firms, a higher level of corporate social committee independence is associated with more social sustainability reporting, whereas a higher different level of gender diversity is affiliated with less social sustainability reporting.

With the low level of sustainability reporting in SSA, which emanates from the poor characteristics of the social sustainability committees of listed businesses in the region, strong regulatory and legal corporate governance framework in the region is essential. It is important for policymakers and regulators of corporate governance in the sub-Saharan African region to establish and implement a single legal and regulatory framework to guide sustainability reporting. The country-level legal and regulatory frameworks on corporate governance in SSA need to be mandatory, regularly revised, and updated to meet emerging framework gaps. The regulators and policymakers in SSA need to strengthen enforcement mechanisms to ensure compliance with corporate governance and sustainability reporting standards. This includes regular audits, penalties for non-compliance, and public disclosure of compliance status. There is also the need for institutional capacity enhancement with the region. Regulators and policymakers in SSA need to strategically invest in training and capacity-building for regulatory bodies to effectively monitor and enforce corporate governance and sustainability reporting standards. Regulators and policymakers would also need to provide resources and training for companies, especially small and medium-sized enterprises (SMEs), to help them understand and implement good corporate governance practices and sustainability reporting. The regulators and policymakers in SSA need to also foster stakeholder engagement and strengthen regional cooperation in the areas of corporate governance framework building and strengthening. There is the need for the promotion of regular dialogues between regulators, companies, investors, civil society, and other stakeholders to discuss challenges and opportunities in corporate governance and sustainability reporting. There is also the need to work with regional bodies such as the African Union and regional economic communities to harmonize corporate governance and sustainability reporting standards across the region, as well as the need to facilitate knowledge sharing and best practice exchanges among countries in sub-Saharan Africa to foster a collaborative approach to improving corporate governance.

6. Conclusions

The social sustainability reporting proportion of 39 percent is relatively lower than the 53.5 percent in developed countries (Bhatia and Makkar 2020). Except in South Africa, social sustainability reporting is largely voluntary, unstandardized, or unregulated at the national level in sub-Saharan Africa. This finding implies that all stock exchanges in the sub-Saharan African region should follow the South African example by also making mandatory the production of a comprehensive integrated sustainability report that incorporates social, economic, and environmental information. Policymakers like legislative bodies in sub-Saharan Africa can standardize, regulate, and mandate comprehensive sustainability reporting in sub-Saharan Africa.

In sub-Saharan Africa, some traits of the board committees of publicly traded companies are critical to the degree of sustainability reporting or disclosure. Effective audit committee size, corporate social committee size, and independence have the potential to

enhance or stimulate a higher level of social sustainability reporting. This implies that regulatory bodies and policymakers should therefore ensure that all publicly listed firms have an efficient membership size of four and also at least one independent member to seek the interest of the public. The establishment of independent board committees with gender diversity is particularly important in institutional investment firms, since institutional shareholding moderates the effect of corporate social committee independence and gender diversity on social sustainability reporting. Regulatory bodies should also ensure that all publicly listed firms have established corporate social committees to design and implement corporate sustainability strategies, including social sustainability reporting.

7. Limitations and Areas for Further Studies

With the reliance of this study on secondary and predetermined data, the researchers' selection of variables was limited or defined by the primary source of this study. Although the authors would have wished to expand the scope of this study to include other areas of corporate governance characteristics like remuneration and nomination committee characteristics, the original data conceptually focused on audit committee characteristics and corporate social sustainability committee characteristics and sustainability reporting. Thus, the limitation of the researcher to the variables defined by the primary source of the data led to the exclusion of nomination committee characteristics as corporate governance characteristics. Future studies are therefore recommended to explore the role of nomination and remuneration committee characteristics in the level of disclosure of sustainability in SSA.

Additionally, this study was restricted to sub-Saharan African nations whose companies had stock market listings between 2012 and 2021. This suggests that this study did not include any sub-Saharan African nations with stock markets or those that created their stock markets after 2021. Future studies can therefore explore the situation of firms listed after 2021, since there is continuous evolution in the legal and regulatory framework on corporate governance at country level and within the sub-Saharan African region. This study also focused on only listed non-financial companies in SSA due to corporate governance and sustainability disclosure metric differences between financial and non-financial businesses. Future studies can therefore replicate this study using listed financial companies in SSA and compare the findings to the current study.

Notwithstanding that there are many countries in sub-Saharan Africa with young stock exchanges or without stock exchanges, the emphasis of this study on publicly listed businesses implies that the defined scope is representation enough to produce significant and representative findings.

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